

Precision, accuracy and uncertainty

Working scientifically
Both papers · Combined and Separate
10 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

	QUESTION	ANSWER	✓
1	Define precision.	How close repeated measurements are to each other.	☐
2	Define accuracy.	How close a measurement is to the true value.	☐
3	Define the resolution of a measuring instrument.	The smallest change the instrument can detect.	☐
4	Define uncertainty.	An estimate of the doubt attached to a measurement.	☐
5	State how the mean of a set of repeated readings is calculated.	Add up the readings, then divide by the number of readings.	☐
6	State how the range of a set of readings is calculated.	Largest reading – smallest reading	☐
7	State how uncertainty is calculated from a set of repeated readings.	Uncertainty = range ÷ 2 (range = largest reading – smallest reading)	☐
8	Describe how a final result should be written when its uncertainty is known.	As mean ± uncertainty, with a unit, e.g. 1.22 ± 0.03 s.	☐
9	Three readings are close together but far from the true value. Describe them using the words precise and accurate.	They are precise but not accurate.	☐
10	State what repeated measurements help you to judge: precision or accuracy.	Precision	☐